

EE2213 Introduction to AI

Lecture 10: Clustering

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OVERVIEW OF COURSE CONTENTS

- **Introduction (Shaojing)**

- What is AI
- Applications of AI
- AI agent

- **Search (Shaojing)**

- Uninformed search algorithms: breadth-first, depth-first, uniform-cost(Dijkstra's algorithm)
- Informed search algorithms: greedy best-first, A*
- Applications

- **Optimisation (Shaojing)**

- Linear programming
- Convex problems
- Applications

- **Machine learning (Wang Si) (Weeks 6-9)**

- Supervised and unsupervised learning: regression, classification, clustering
- Neural networks and deep learning
- Applications

- **Knowledge representation (Wang Si) (Weeks 10-11)**

- Knowledge Representation and Reasoning
- Propositional Logic
- Applications

- **Ethical considerations (Shaojing)**

- Bias in AI
- Privacy concerns
- Societal impact

(Week 10, Monday)

**No Class on 20 Oct.! A recording
will be uploaded on Canvas.**

AGENDA

- We will discuss:
 - types of clusters and types of clustering
 - K-means clustering
 - Fuzzy C-means clustering

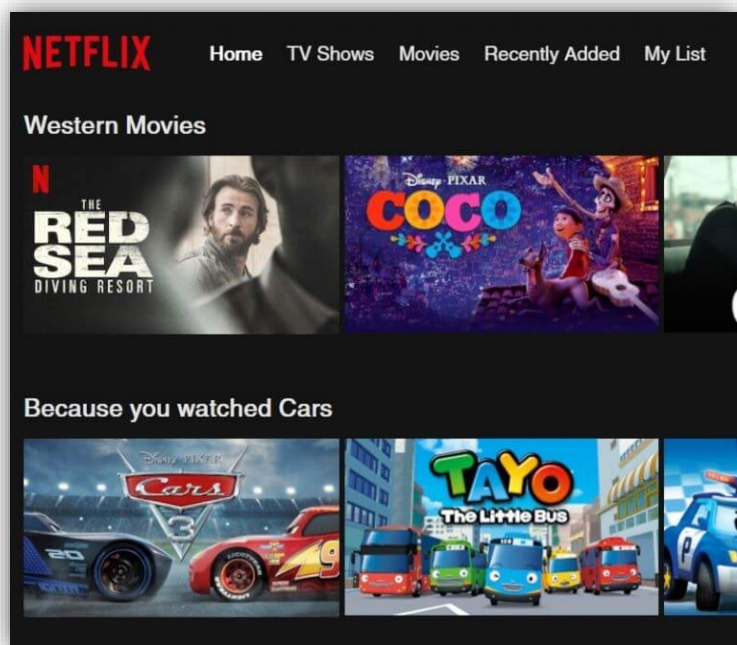
At the end of this lecture, you should be able to:

- ✓ Differentiate between types of clusters and clustering
- ✓ Apply K-means and Fuzzy C-means
- ✓ Describe the limitations of K-means clustering

Recap: Clustering

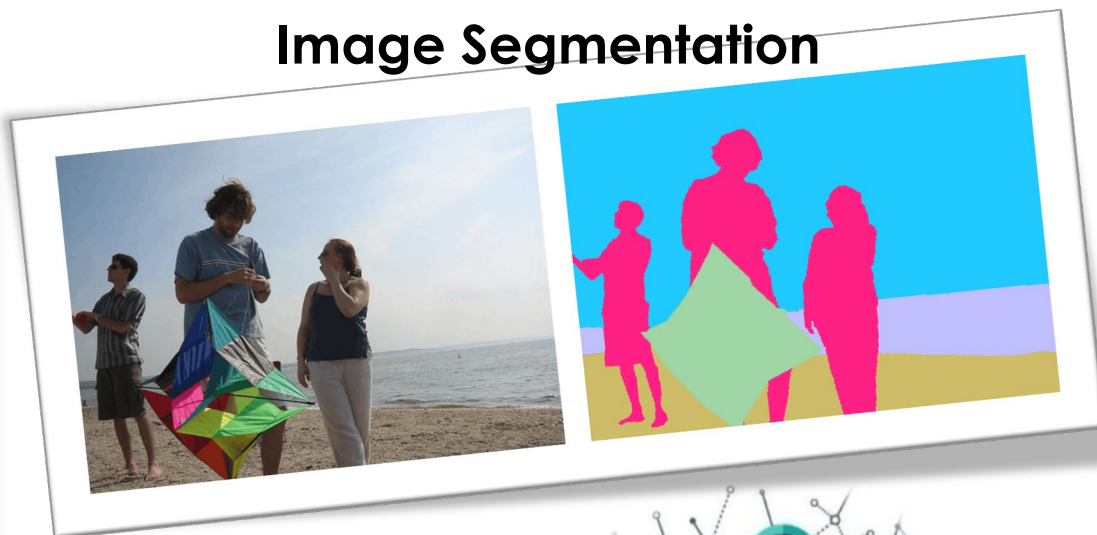
Groups a set of objects in such a way that objects in the same group (called a **cluster**) are **more similar** (in some sense) to each other than to those in other groups (clusters).

Applications:



Recommender systems

Image Segmentation



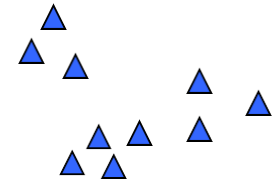
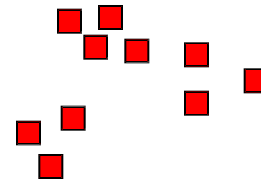
Social
Network
Analysis



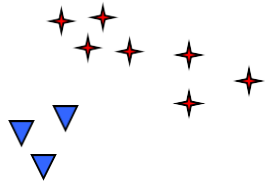
Notion of a Cluster can be Ambiguous



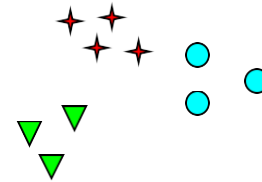
How many clusters?



Two Clusters



Four Clusters



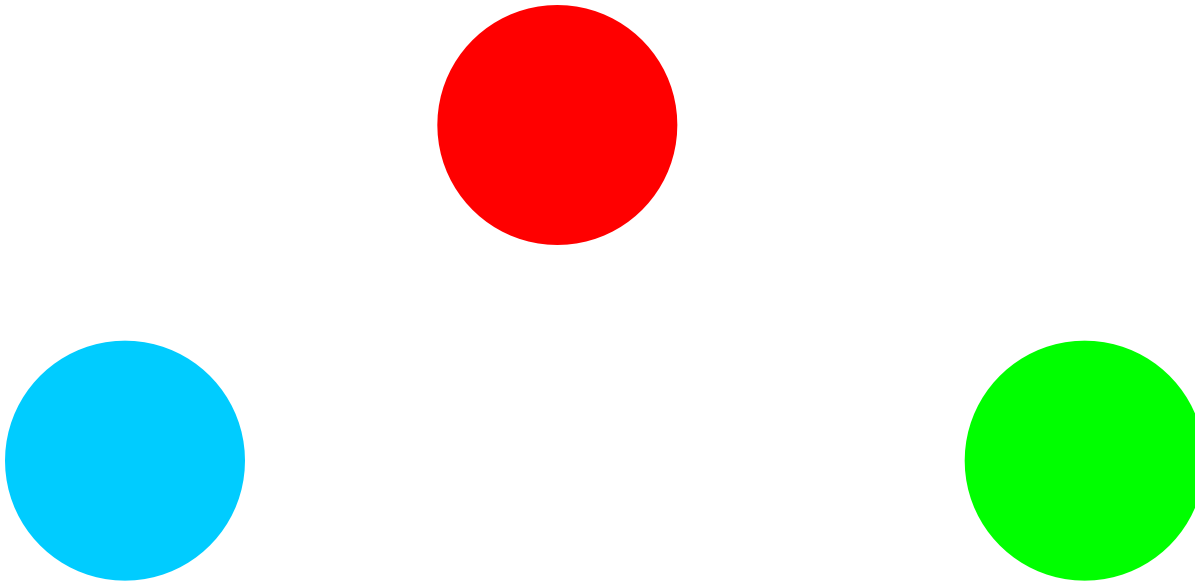
Six Clusters

Types of Clusters

- Well-separated Clusters
- Center-based Clusters
- Contiguous Clusters
- Density-based Clusters

Types of Clusters

- Well-Separated Cluster:
 - A cluster is a set of points such that any point in a cluster is closer (or more similar) to every other point in the cluster than to any point not in the cluster.

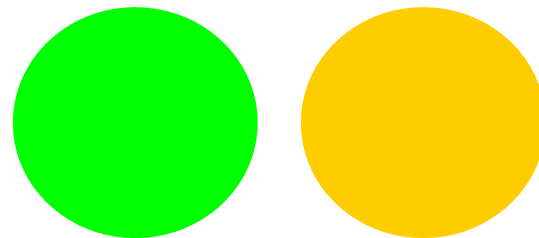
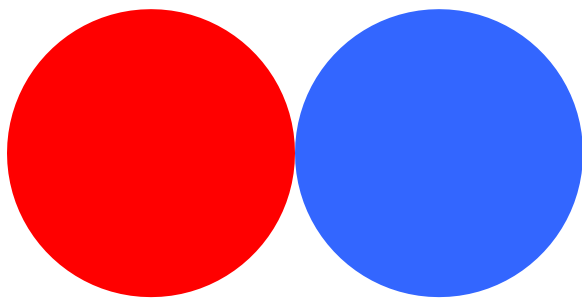


3 well-separated clusters

Types of Clusters

- Center-based Cluster:

- A cluster is a set of objects such that a point in a cluster is closer (more similar) to the “center” of a cluster, than to the center of any other cluster
- The center of a cluster is often a centroid, the average of all the points in the cluster, or the most “representative” point of a cluster



4 center-based clusters

How many well-separated clusters?

Types of Clusters

- Contiguous Cluster (Nearest neighbor)
 - A cluster is a set of points such that a point in a cluster is closer (or more similar) to **one or more** other points in the cluster than to any point not in the cluster.



8 contiguous clusters

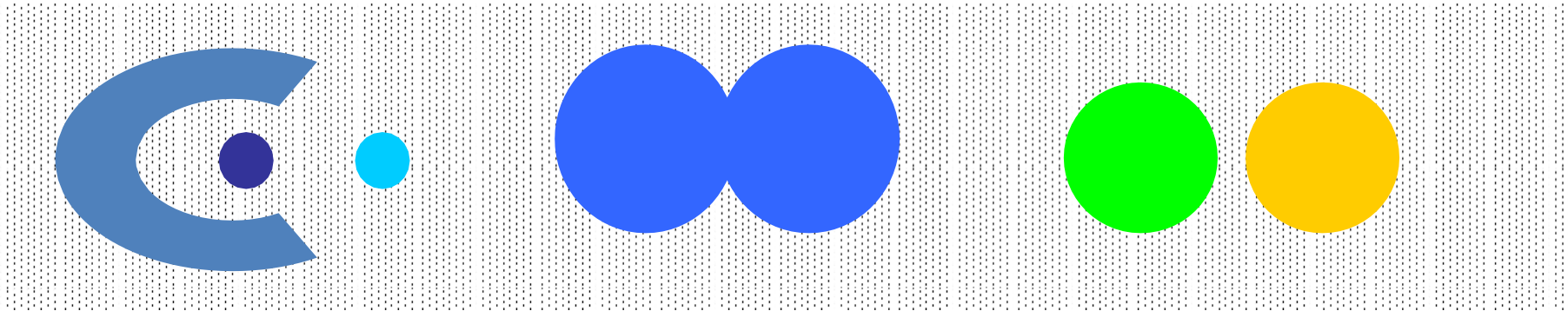
How many well-separated clusters?

How many center-based clusters?

Types of Clusters

- Density-based Cluster

- A cluster is a dense region of points, which is separated by low- density regions, from other regions of high density.
- Used when noise and outliers are present.



6 density-based clusters

Types of Clustering

- Hard Clustering
 - Each data point belongs to a cluster completely or not.
 - e.g., **K-means**, Hierarchical, etc.
- Soft Clustering
 - Each data point belongs to each cluster with a probability or degree of membership.
 - e.g., **Fuzzy C-means**, Gaussian mixture model, etc.

Data Points	Hard	Soft	
	Clusters	P(C1)	P(C2)
A	C1	0.91	0.09
B	C2	0.3	0.7
C	C2	0.17	0.83
D	C1	1	0



K-means Clustering

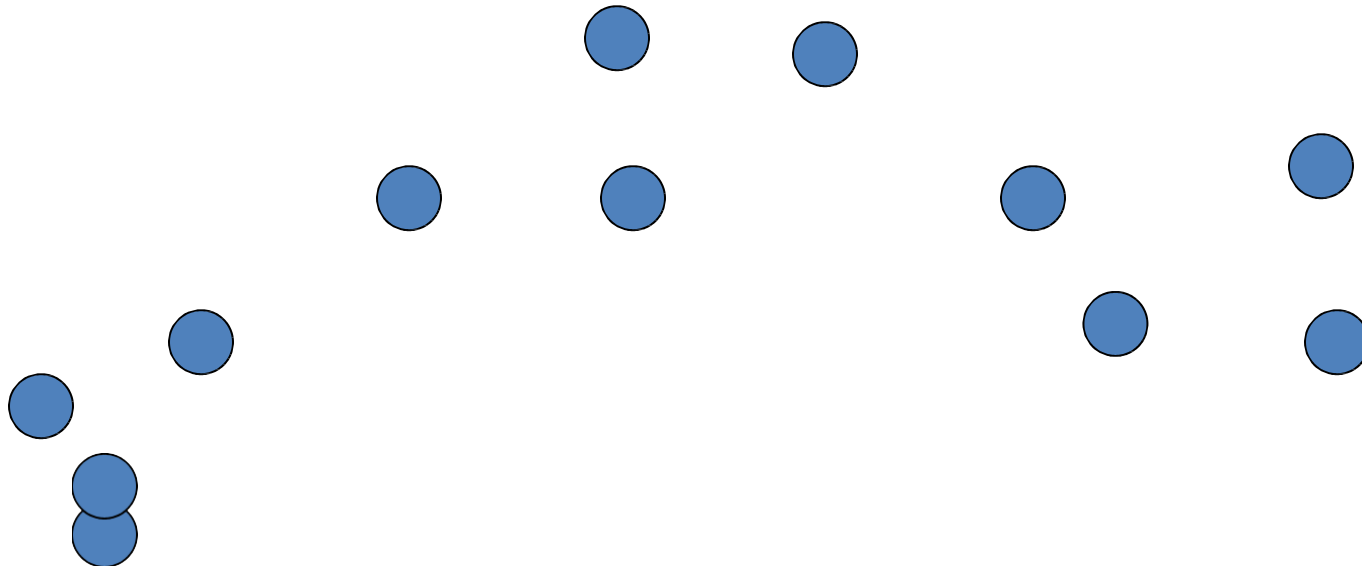
-Hard Clustering

What type of
clusters?

K-means Clustering

- ✓ Each cluster is associated with a centroid (center point)
- ✓ Each point is assigned to the cluster with the closest centroid

An example: $K=3$ (we seek 3 clusters)

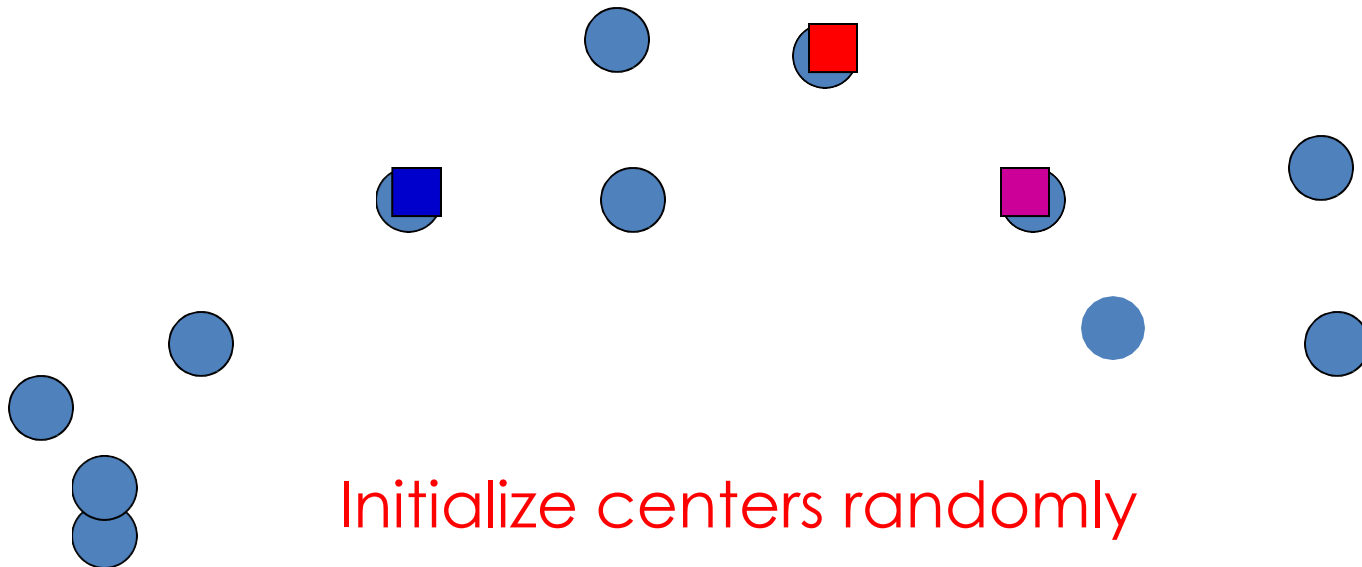


K-means Clustering

- ✓ Each cluster is associated with a centroid (center point)
- ✓ Each point is assigned to the cluster with the closest centroid

An example: $K=3$ (we seek 3 clusters)

Step 1: Randomly choose 3 points as the centers of the 3 clusters



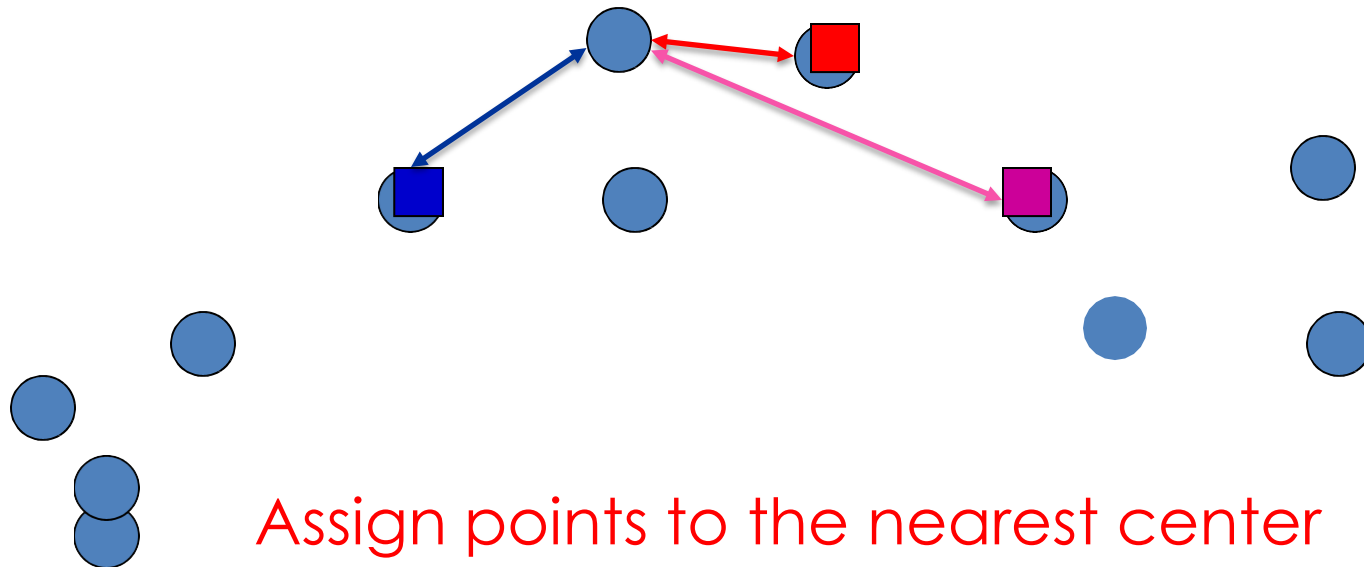
Initialize centers randomly

K-means Clustering

- ✓ Each cluster is associated with a centroid (center point)
- ✓ Each point is assigned to the cluster with the closest centroid

An example: $K=3$ (we seek 3 clusters)

Step 2: For each point, calculate the distance between the point and the 3 centers, and assign it to the nearest one.

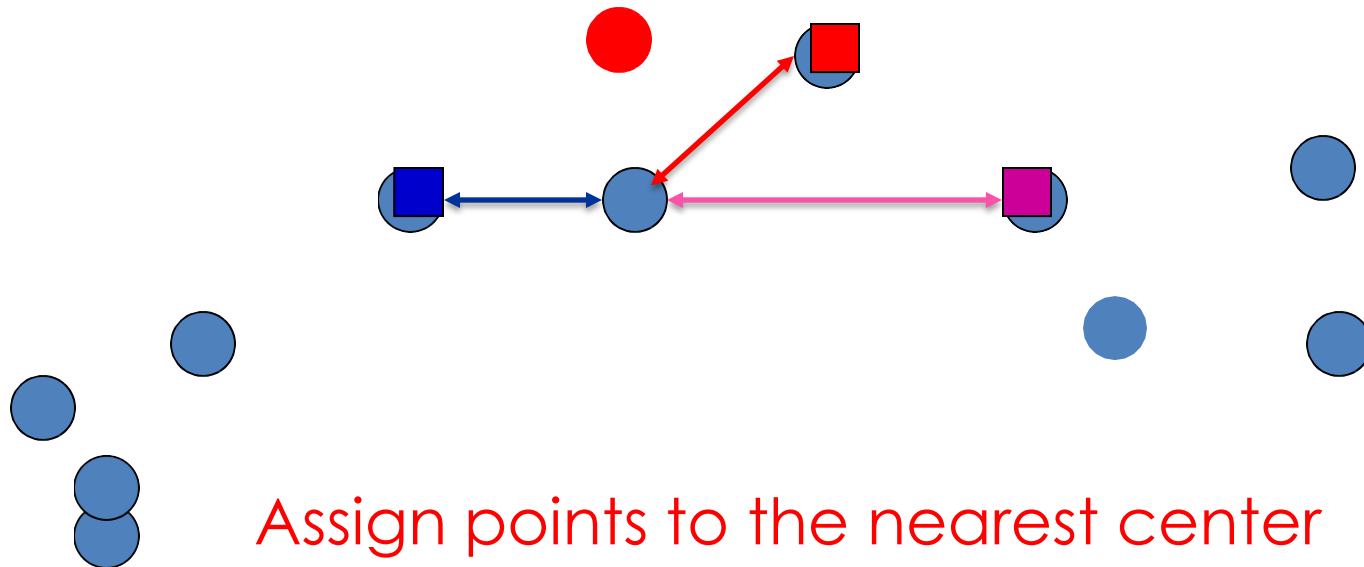


K-means Clustering

- ✓ Each cluster is associated with a centroid (center point)
- ✓ Each point is assigned to the cluster with the closest centroid

An example: $K=3$ (we seek 3 clusters)

Step 2: For each point, calculate the distance between the point and the 3 centers, and assign it to the nearest one.



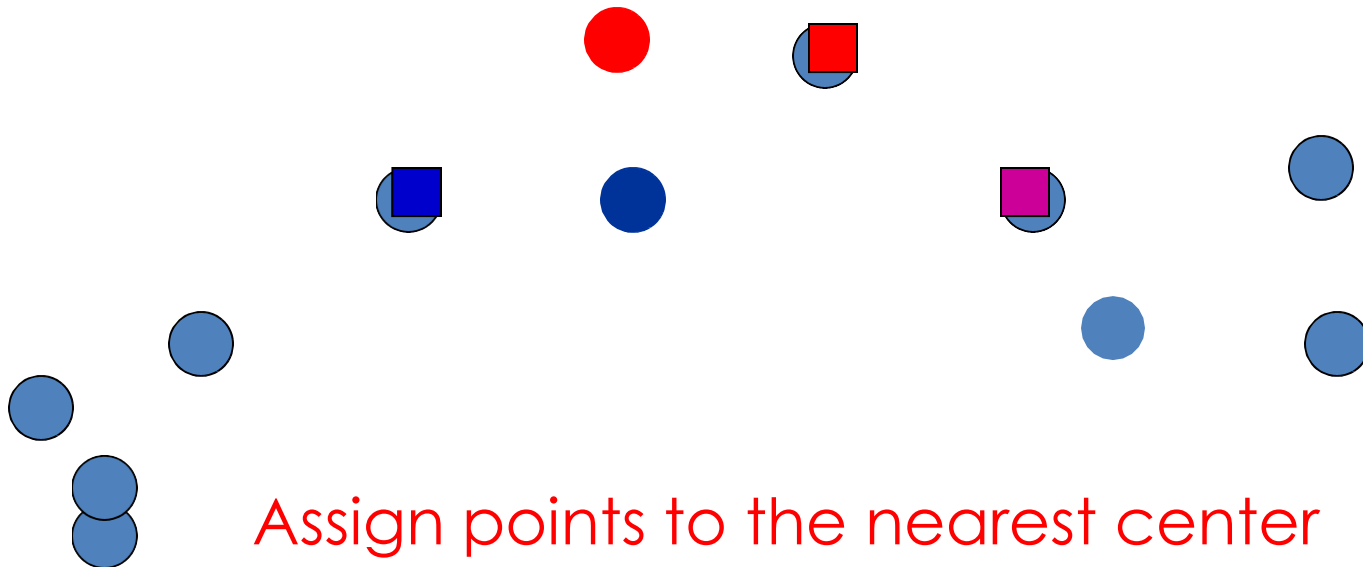
Assign points to the nearest center

K-means Clustering

- ✓ Each cluster is associated with a centroid (center point)
- ✓ Each point is assigned to the cluster with the closest centroid

An example: $K=3$ (we seek 3 clusters)

Step 2: For each point, calculate the distance between the point and the 3 centers, and assign it to the nearest one.

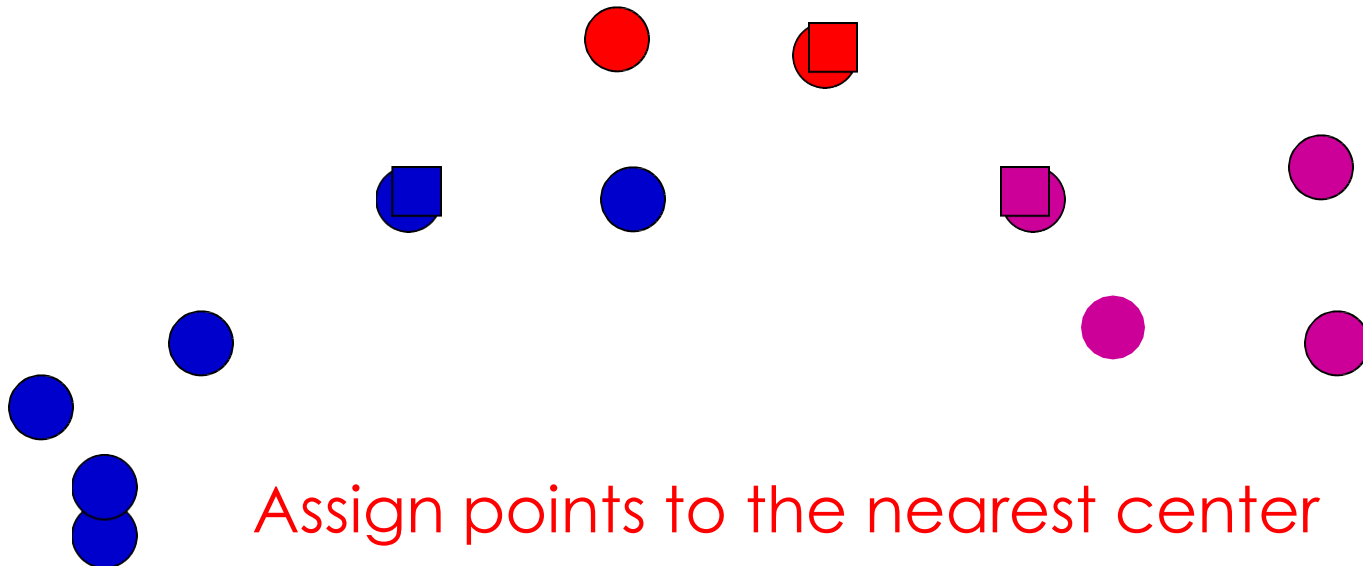


K-means Clustering

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An example: $K=3$ (we seek 3 clusters)

Step 2: For each point, calculate the distance between the point and the 3 centers, and assign it to the nearest one.

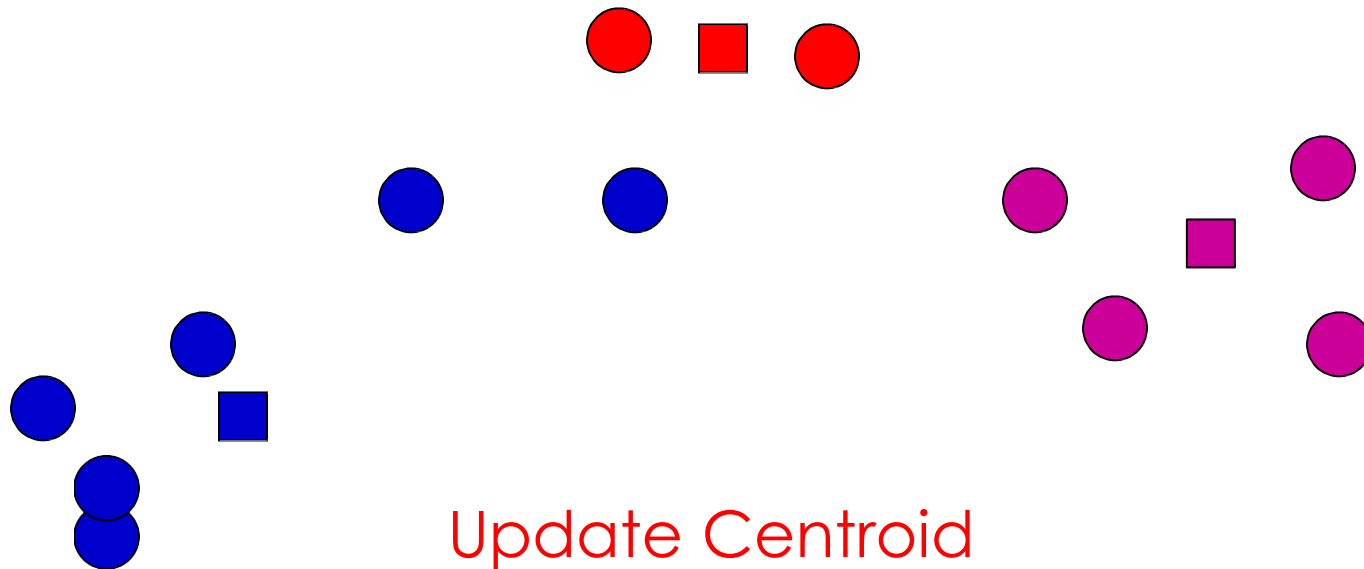


K-means Clustering

- ✓ Each cluster is associated with a centroid (center point)
- ✓ Each point is assigned to the cluster with the closest centroid

An example: $K=3$ (we seek 3 clusters)

Step 3: Recompute the 3 centers given the current cluster assignments

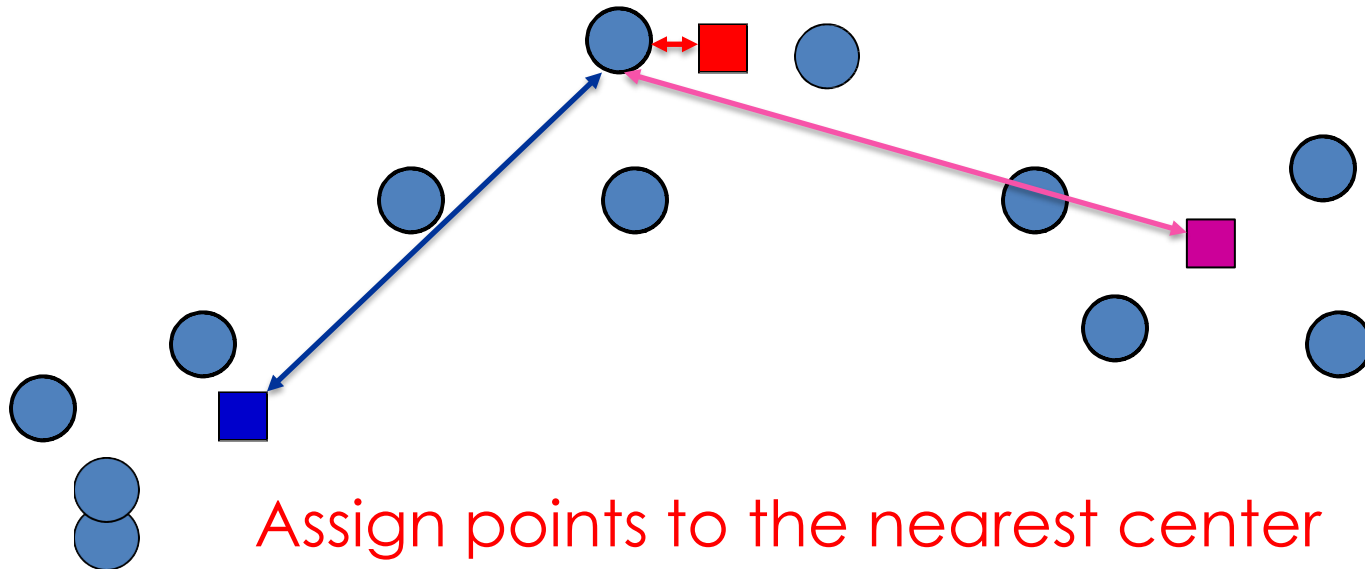


K-means Clustering

- ✓ Each cluster is associated with a centroid (center point)
- ✓ Each point is assigned to the cluster with the closest centroid

An example: $K=3$ (we seek 3 clusters)

Step 4 (repeat Step 2): For each point, calculate the distance between the point and the updated 3 centers, and assign it to the nearest one.

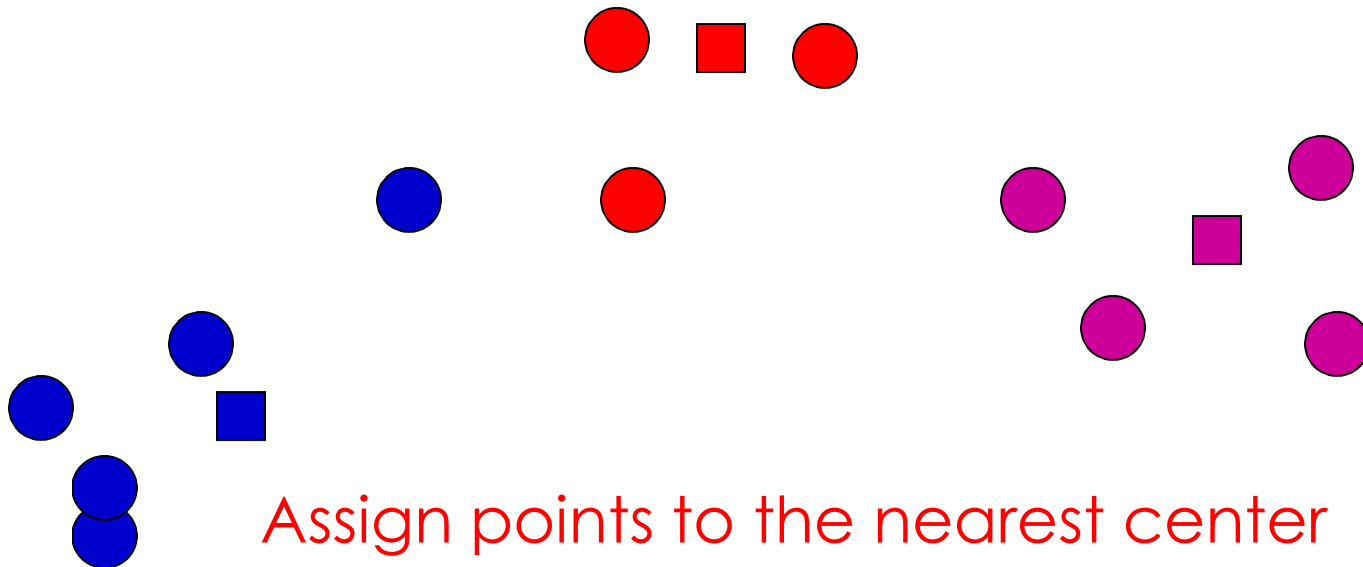


K-means Clustering

- ✓ Each cluster is associated with a centroid (center point)
- ✓ Each point is assigned to the cluster with the closest centroid

An example: $K=3$ (we seek 3 clusters)

Step 4 (repeat Step 2): For each point, calculate the distance between the point and the updated 3 centers, and assign it to the nearest one.

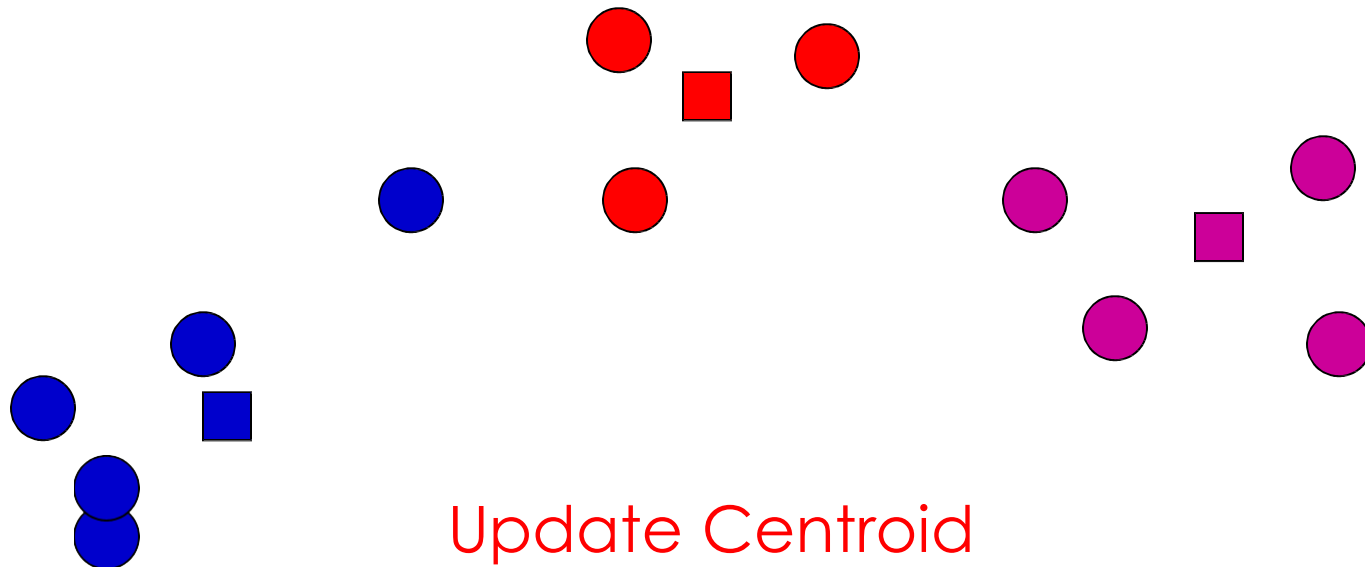


K-means Clustering

- ✓ Each cluster is associated with a centroid (center point)
- ✓ Each point is assigned to the cluster with the closest centroid

An example: $K=3$ (we seek 3 clusters)

Step 5 (repeat Step 3): Recompute the 3 centers given the current cluster assignments.

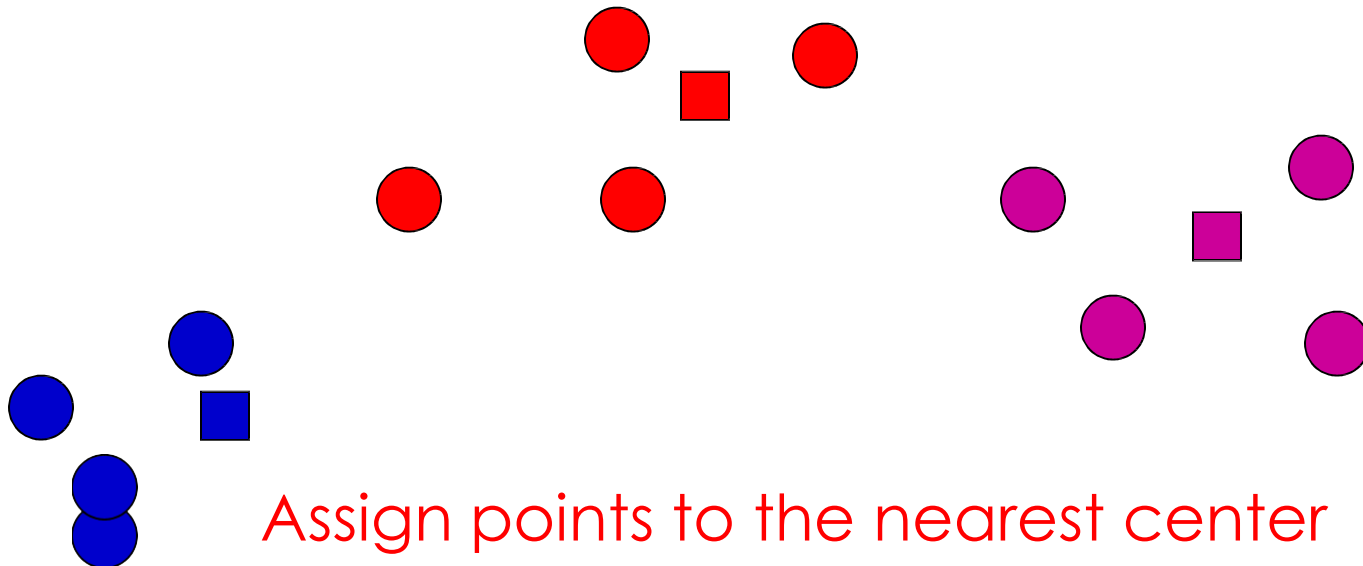


K-means Clustering

- ✓ Each cluster is associated with a centroid (center point)
- ✓ Each point is assigned to the cluster with the closest centroid

An example: $K=3$ (we seek 3 clusters)

Step 6 (repeat Step 2): For each point, calculate the distance between the point and the updated 3 centers, and assign it to the nearest one.

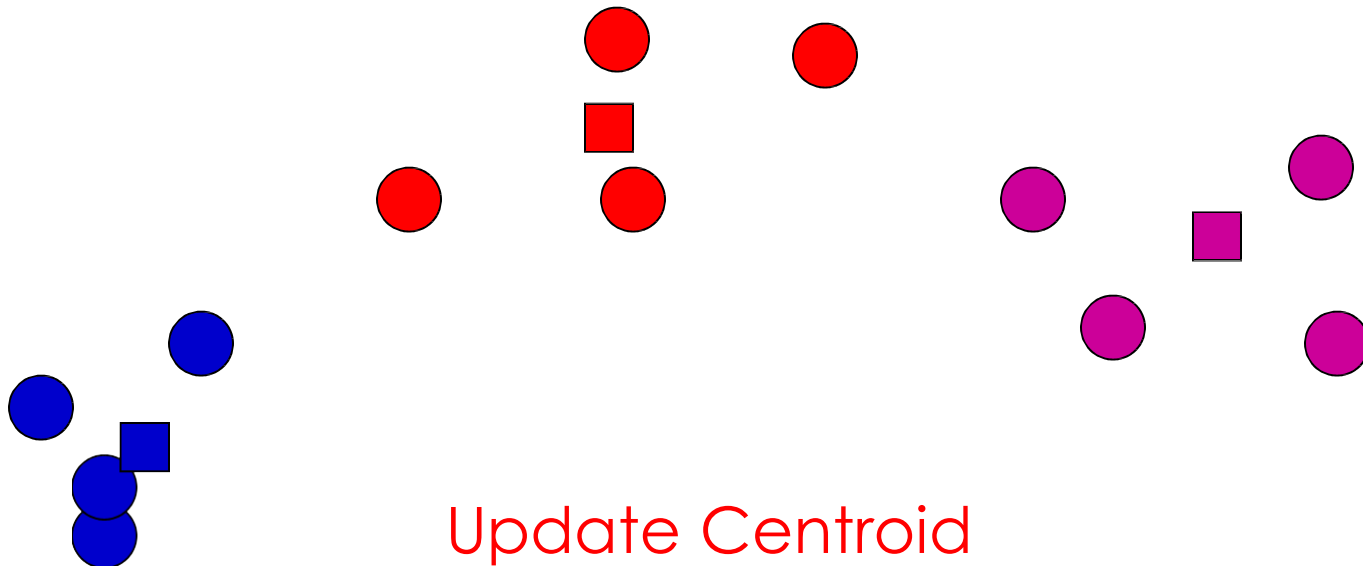


K-means Clustering

- ✓ Each cluster is associated with a centroid (center point)
- ✓ Each point is assigned to the cluster with the closest centroid

An example: $K=3$ (we seek 3 clusters)

Step 7 (repeat Step 3): Recompute the 3 centers given the current cluster assignments.

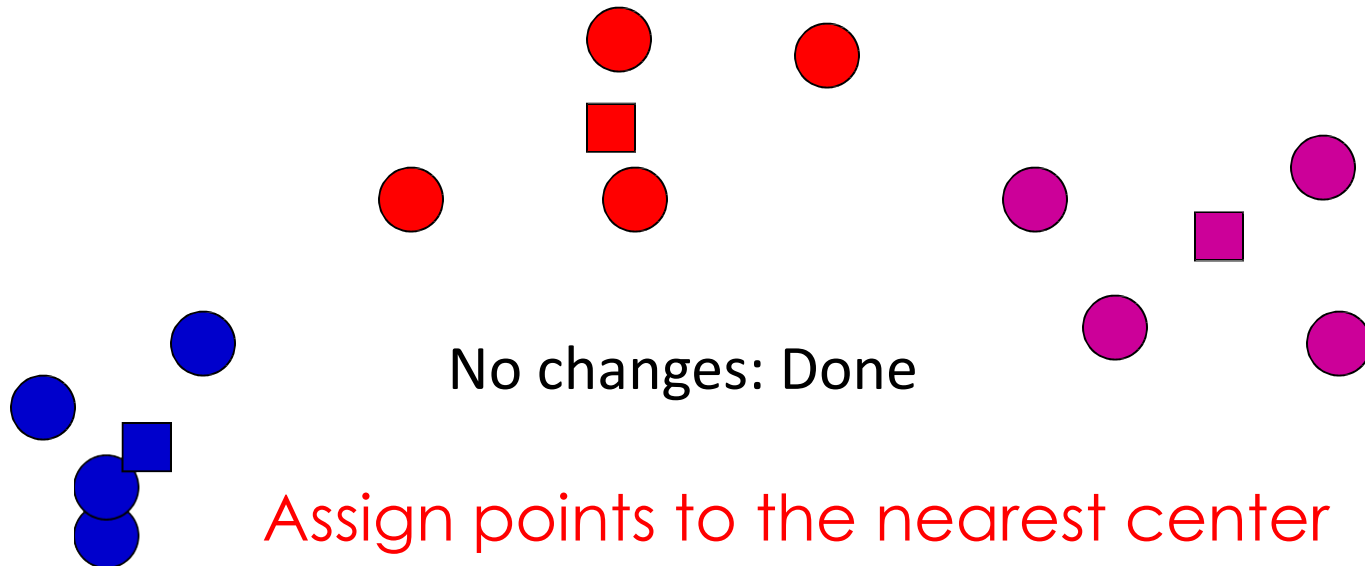


K-means Clustering

- ✓ Each cluster is associated with a centroid (center point)
- ✓ Each point is assigned to the cluster with the closest centroid

An example: $K=3$ (we seek 3 clusters)

Step 8 (repeat Step 2): For each point, calculate the distance between the point and the updated 3 centers, and assign it to the nearest one.



K-means Clustering

➤ Algorithm

Looping between **Assignment** and **Centroid Update**

- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-

K-means Clustering

➤ Objective Function (within-cluster variance)

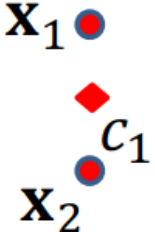
Find the optimal values for $\mathbf{W}_{N \times K}^*$ and $\mathbf{C}_{K \times d}^*$ which minimizes:

$$J(\mathbf{W}, \mathbf{C}) = \sum_{i=1}^N \sum_{k=1}^K w_{ik} \boxed{\|\mathbf{x}_i - \mathbf{c}_k\|^2} \quad \text{Squared Euclidean distance}$$

where $w_{ik} \in \{0,1\}$: $\begin{cases} w_{ik} = 1, & \text{if } \mathbf{x}_i \in \text{cluster } S_k \\ w_{ik} = 0, & \text{else} \end{cases}$

$\mathbf{c}_k$ (d -dimensional vector) is the centroid of cluster S_k

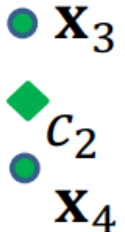
e.g.,



$\mathbf{x}_1$

$\mathbf{x}_2$

$\mathbf{c}_1$



$\mathbf{x}_3$

$\mathbf{x}_4$

$\mathbf{c}_2$

$w_{11} = ?$

$w_{31} = ?$

$w_{12} = ?$

$w_{32} = ?$

$w_{21} = ?$

$w_{41} = ?$

$w_{22} = ?$

$w_{42} = ?$

$J(\mathbf{W}, \mathbf{C}) = ?$

K-means Clustering

➤ Formal Formulation

1. Assignment Step (Fix $\mathbf{C}_{K \times d}$ and update $\mathbf{W}_{N \times K}$)

$$\text{If } \|\mathbf{x}_i - \mathbf{c}_k\|^2 < \|\mathbf{x}_i - \mathbf{c}_{\forall j \neq k}\|^2 : w_{ik} = 1 \\ (\text{else } w_{ik} = 0) \quad i = 1, \dots, N; j, k = 1, \dots, K$$

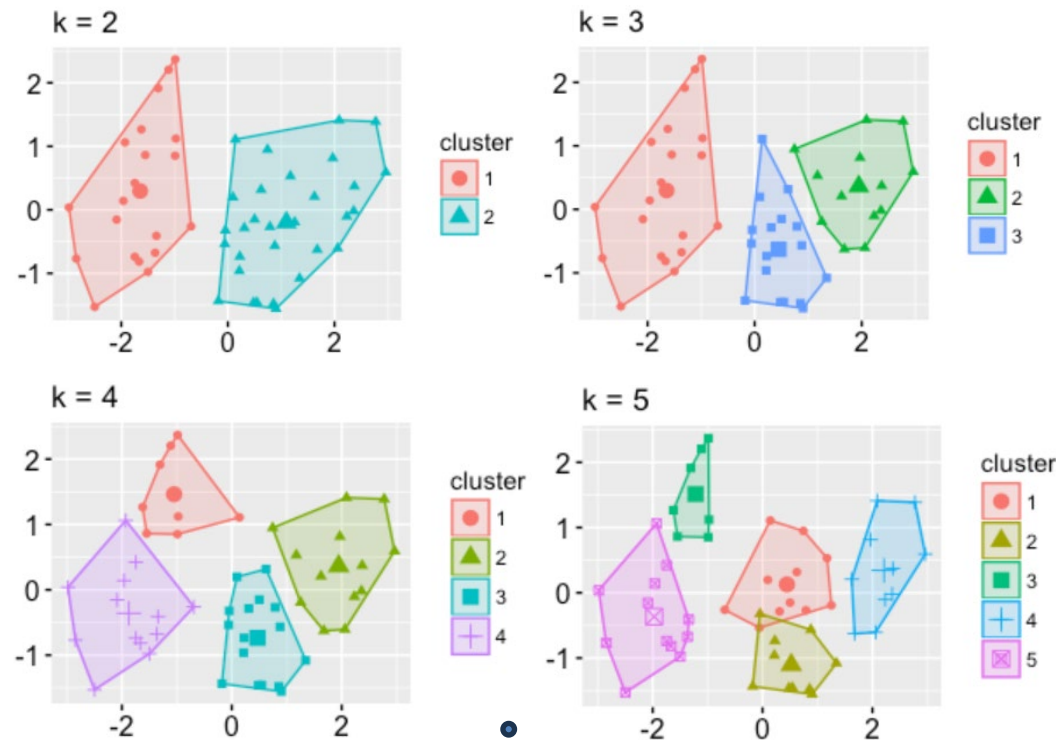
2. Update Step (Fix $\mathbf{W}_{N \times K}$ and update $\mathbf{C}_{K \times d}$)

$$\frac{dJ(\mathbf{W}, \mathbf{C})}{d\mathbf{C}} = \begin{bmatrix} \frac{\partial \sum_{i=1}^N \sum_{k=1}^K w_{ik} (\mathbf{x}_i - \mathbf{c}_k)^T (\mathbf{x}_i - \mathbf{c}_k)}{\partial \mathbf{c}_1} \\ \vdots \\ \frac{\partial \sum_{i=1}^N \sum_{k=1}^K w_{ik} (\mathbf{x}_i - \mathbf{c}_k)^T (\mathbf{x}_i - \mathbf{c}_k)}{\partial \mathbf{c}_K} \end{bmatrix} = \begin{bmatrix} -2 \sum_{i=1}^N w_{i1} (\mathbf{x}_i - \mathbf{c}_1) \\ \vdots \\ -2 \sum_{i=1}^N w_{iK} (\mathbf{x}_i - \mathbf{c}_K) \end{bmatrix} = \mathbf{0}$$

$$\mathbf{c}_k = \frac{\sum_{i=1}^N w_{ik} \mathbf{x}_i}{\sum_{i=1}^N w_{ik}}, k = 1, \dots, K$$

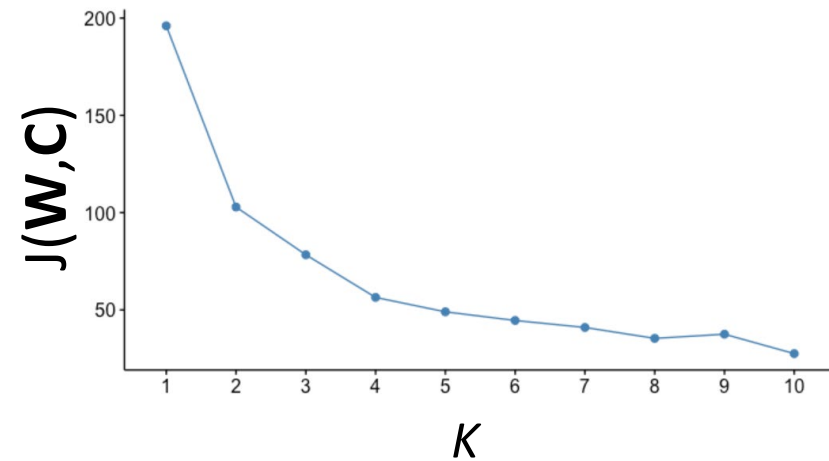
K-means Clustering: Limitations

➤ Need to predefine K



Elbow Method:

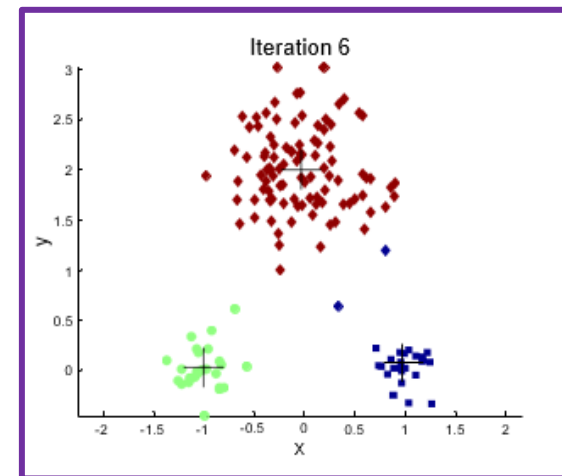
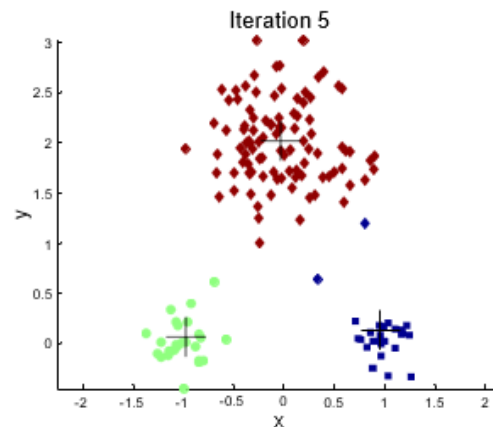
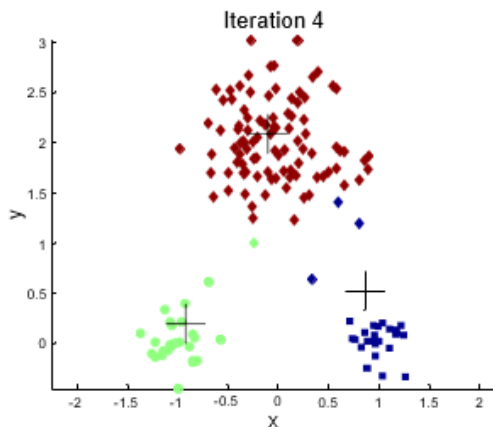
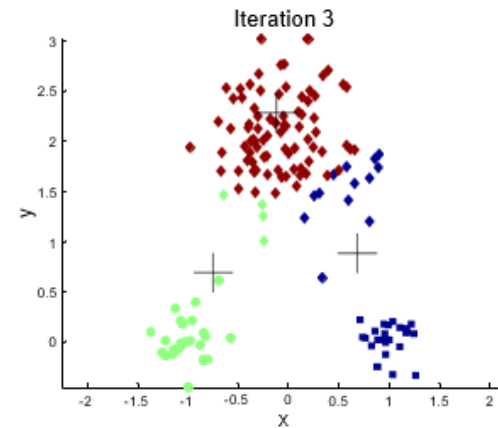
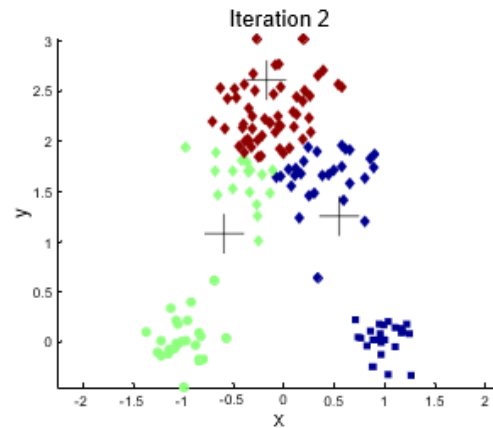
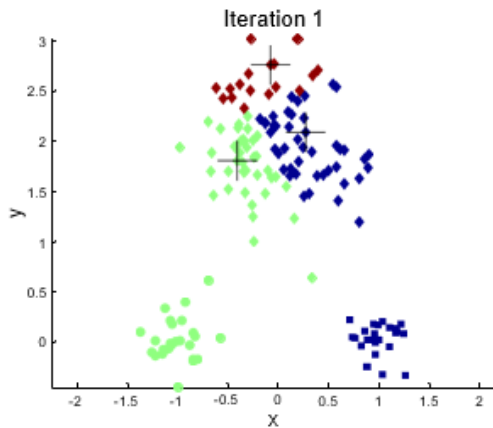
1. K-means for different K
2. For each K , calculate $J(\mathbf{W}, \mathbf{C})$
3. Plot the curve of $J(\mathbf{W}, \mathbf{C})$ over K
4. “bend” or “knee”



What is the optimal K ?

K-means Clustering: Limitations

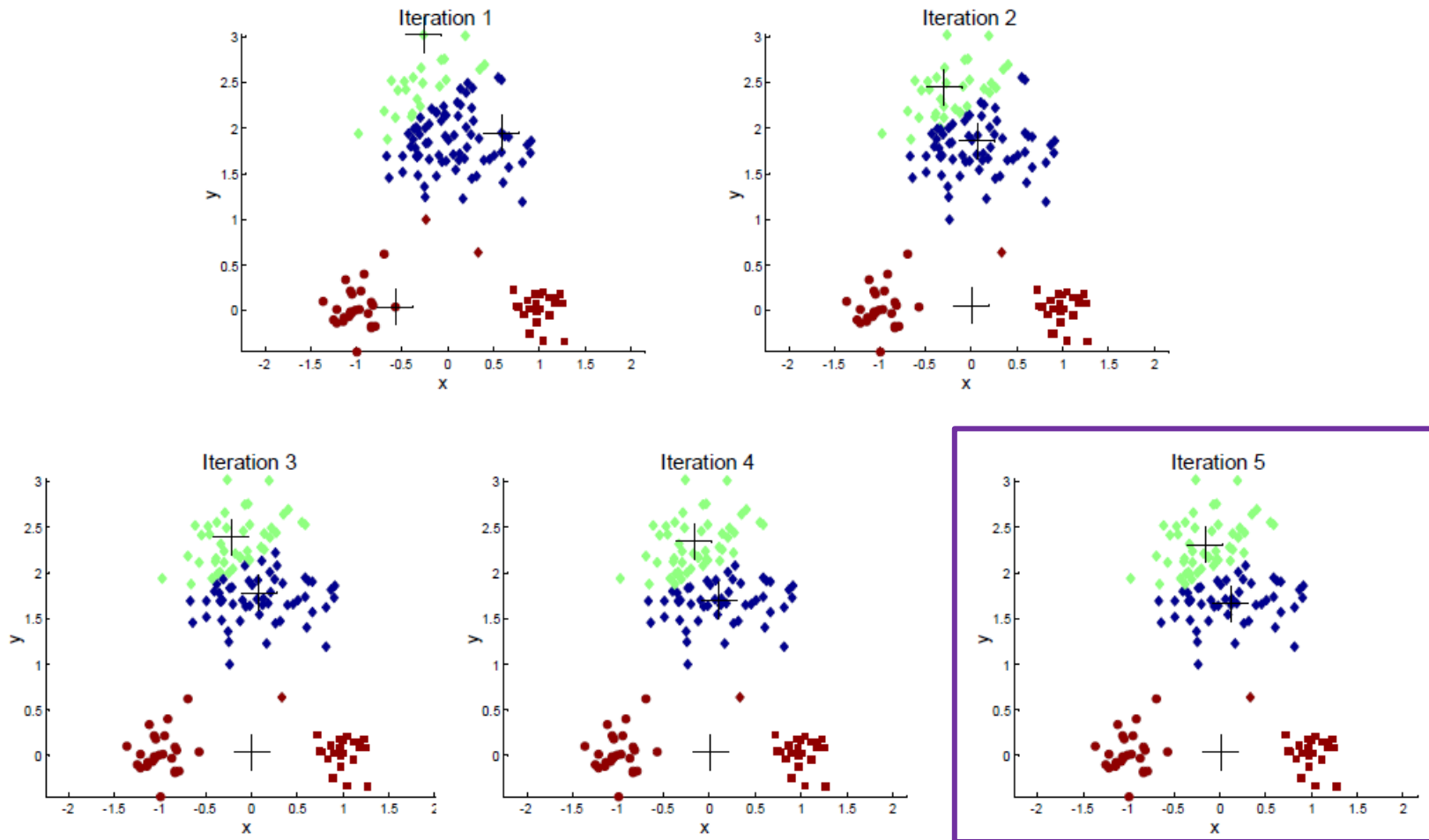
➤ Sensitive to initial centroid selection



Optimal

K-means Clustering: Limitations

➤ *Sensitive to initial centroid selection*

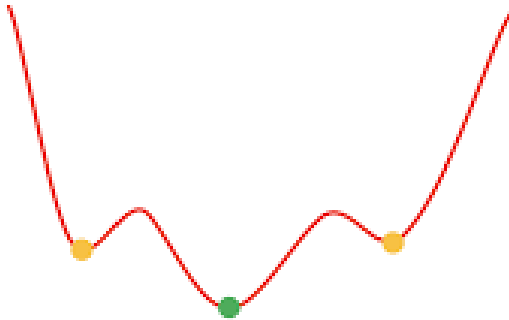


Suboptimal

K-means Clustering: Limitations

➤ *Sensitive to initial centroid selection*

$J(\mathbf{W}, \mathbf{C})$ is non-convex



K-means is not guaranteed to find a global minimum; it finds only local minimum.

How to guarantee a global minimum?

K-means Clustering: Limitations

➤ *Sensitive to initial centroid selection*

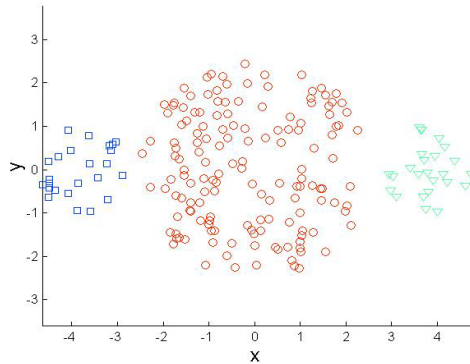
Practically Possible Trials:

- Multiple runs with different random initializations
 - Helps, but probability is not on your side
- Select more than K initial centroids and then select among these initial centroids
 - Select most widely separated
- Post-processing
 - Eliminate small clusters that may represent outliers
 - Split 'loose' clusters, i.e., clusters with relatively high $J(\mathbf{W}, \mathbf{C})$
 - Merge clusters that are 'close' and that have relatively low $J(\mathbf{W}, \mathbf{C})$

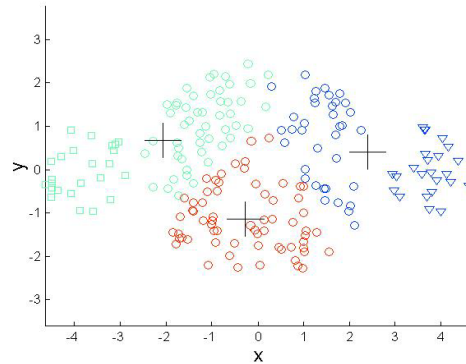
K-Means++: Improved initialization for K-Means

K-means Clustering: Limitations

➤ **Assume spherical, equal-sized, similar-density clusters**

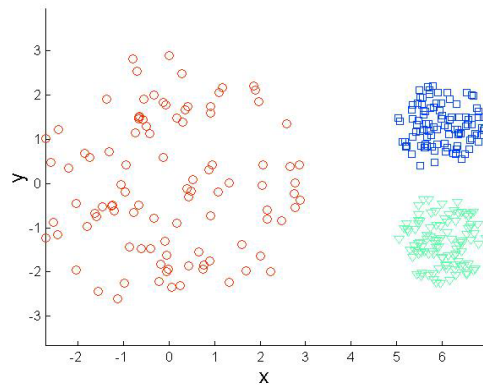


Original Points

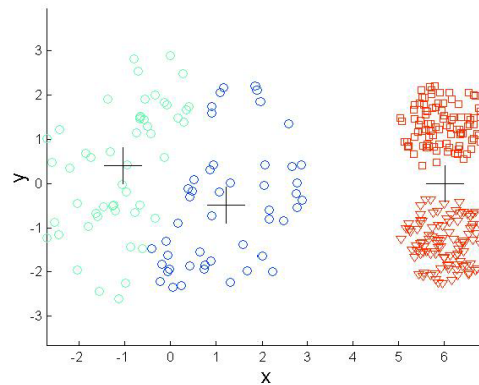


K-means (3 Clusters)

Differ in size!



Original Points



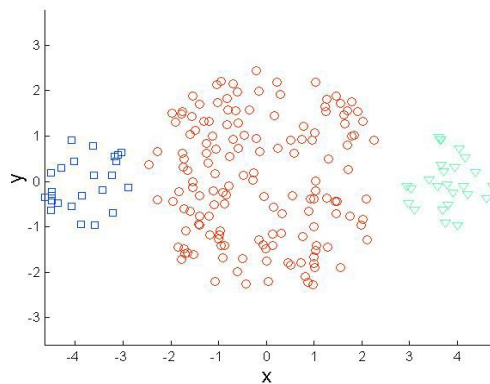
K-means (3 Clusters)

Differ in Density!

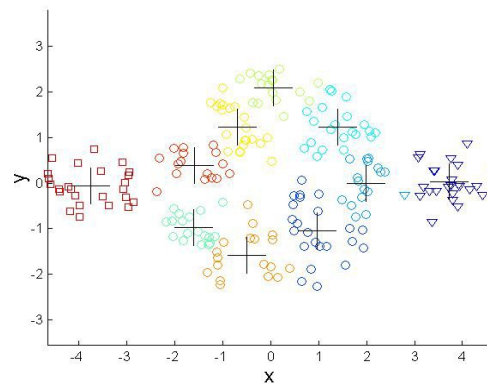
K-means Clustering: Limitations

➤ *Assume spherical, equal-sized, similar-density clusters*

Possible Solution: Use large K , then postprocess

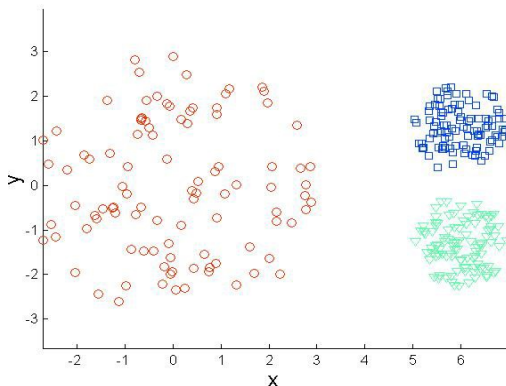


Original Points

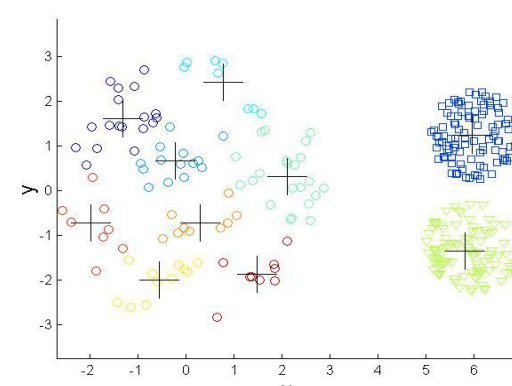


K-means (10 Clusters)

Differ in size



Original Points



K-means (10 Clusters)

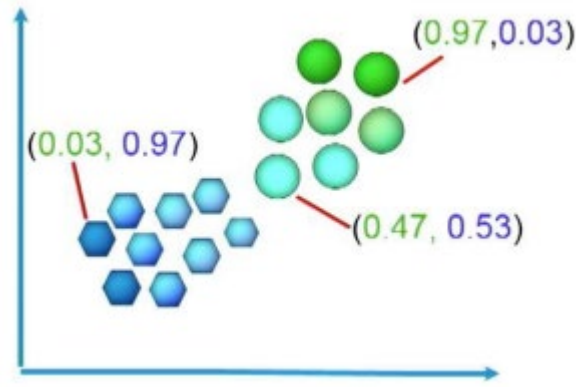
Differ in Density

K-means Clustering: Example

Consider the following data points:

Sample ID	1	2	3	4	5	6
Sample	1	2	3	10	11	12

We'd like to group them into two groups by K-means according to Euclidean distance. Assume the initial centroid is Sample 1 for Group A, and Sample 4 for Group B. Upon convergence, what will be the two centres?



Fuzzy C-Means Clustering

-Soft Clustering

Fuzzy C-Means Clustering

➤ Objective Function

Find the optimal values for $\mathbf{W}_{N \times C}^*$ and $\mathbf{C}_{C \times d}^*$ which minimizes:

$$J(\mathbf{W}, \mathbf{C}) = \sum_{i=1}^N \sum_{k=1}^C (w_{ik})^r \boxed{\|\mathbf{x}_i - \mathbf{c}_k\|^2}$$

Squared
Euclidean
distance

where $w_{ik} \in [0,1]$: $w_{ik} = \frac{1}{\sum_{j=1}^C \left(\frac{\|\mathbf{x}_i - \mathbf{c}_k\|}{\|\mathbf{x}_i - \mathbf{c}_j\|} \right)^{\frac{2}{r-1}}}$ tells the degree of $\mathbf{x}_i$ belonging to cluster $\mathbf{c}_k$

$r > 1$ is the fuzzifier: determines the level of cluster fuzziness.
(usually, $1.25 \leq r \leq 2$)

Fuzzy C-Means Clustering

➤ Algorithm Formulation

1. Assignment Step (Fix $\mathbf{C}_{C \times d}$ and update $\mathbf{W}_{N \times C}$)

$$w_{ik} = \frac{1}{\sum_{j=1}^C \left(\frac{\|\mathbf{x}_i - \mathbf{c}_k\|}{\|\mathbf{x}_i - \mathbf{c}_j\|} \right)^{\frac{2}{r-1}}}$$

2. Update Step (Fix $\mathbf{W}_{N \times C}$ and update $\mathbf{C}_{C \times d}$)

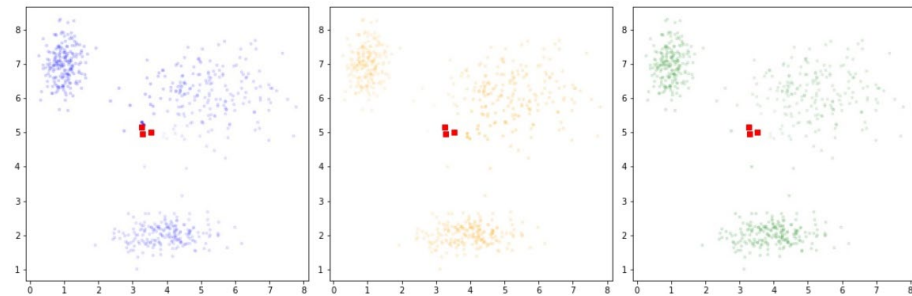
$$\frac{dJ(\mathbf{W}, \mathbf{C})}{d\mathbf{C}} = \begin{bmatrix} \frac{\partial \sum_{i=1}^N \sum_{k=1}^C (w_{ik})^r (\mathbf{x}_i - \mathbf{c}_k)^T (\mathbf{x}_i - \mathbf{c}_k)}{\partial \mathbf{c}_1} \\ \vdots \\ \frac{\partial \sum_{i=1}^N \sum_{k=1}^C (w_{ik})^r (\mathbf{x}_i - \mathbf{c}_k)^T (\mathbf{x}_i - \mathbf{c}_k)}{\partial \mathbf{c}_K} \end{bmatrix} = \begin{bmatrix} -2 \sum_{i=1}^N (w_{i1})^r (\mathbf{x}_i - \mathbf{c}_1) \\ \vdots \\ -2 \sum_{i=1}^N (w_{iC})^r (\mathbf{x}_i - \mathbf{c}_K) \end{bmatrix} = \mathbf{0}$$

$$\mathbf{c}_k = \frac{\sum_{i=1}^N (w_{ik})^r \mathbf{x}_i}{\sum_{i=1}^N (w_{ik})^r}, k = 1, \dots, C$$

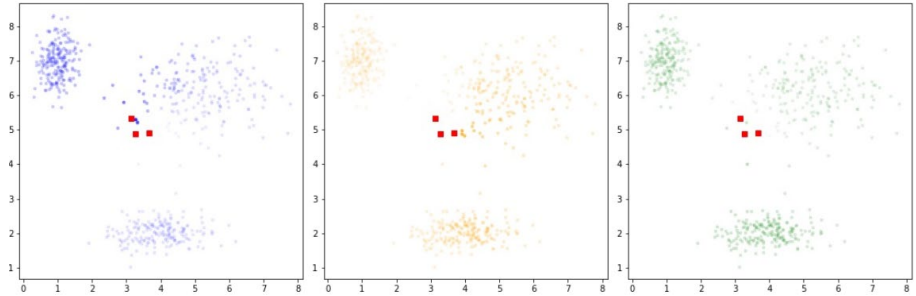
Fuzzy C-Means Clustering

➤ Visualization

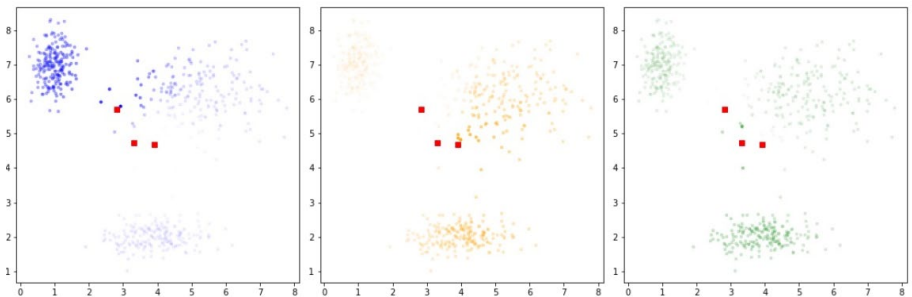
Iteration 1



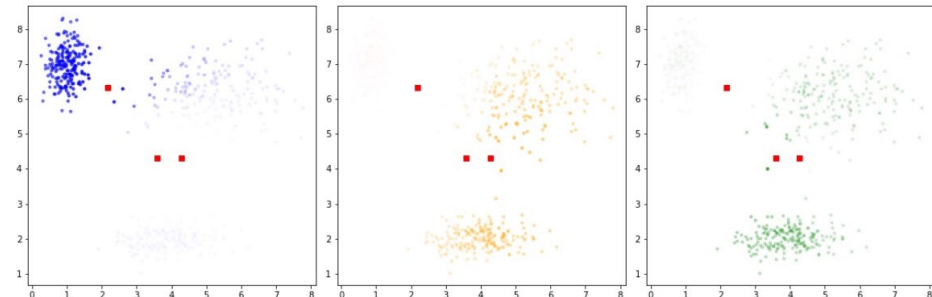
Iteration 2



Iteration 3



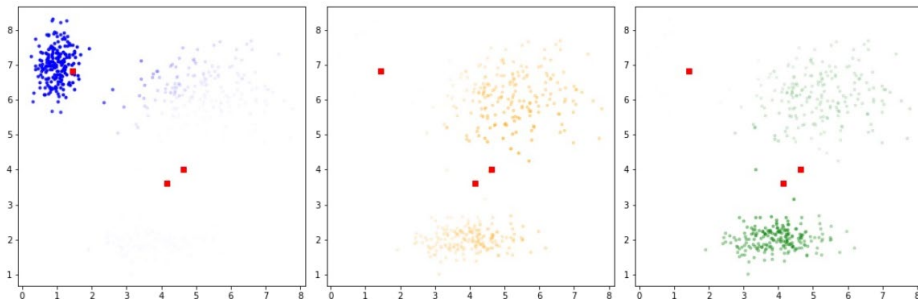
Iteration 4



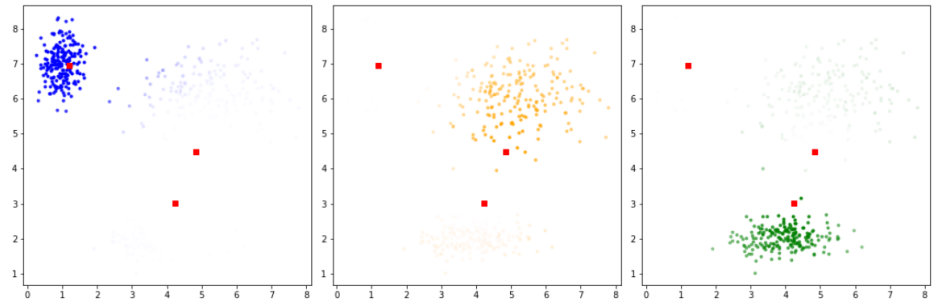
Fuzzy C-Means Clustering

➤ Visualization

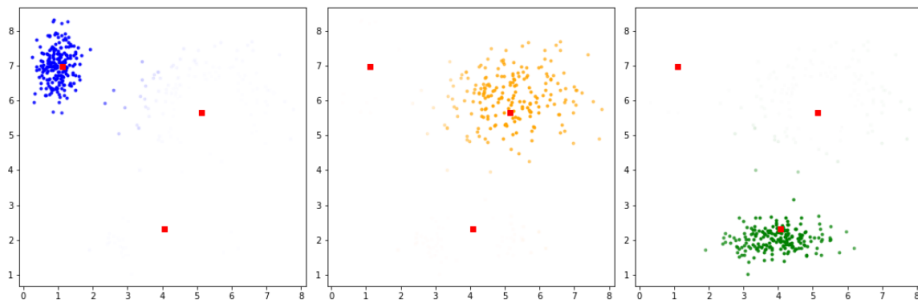
Iteration 5



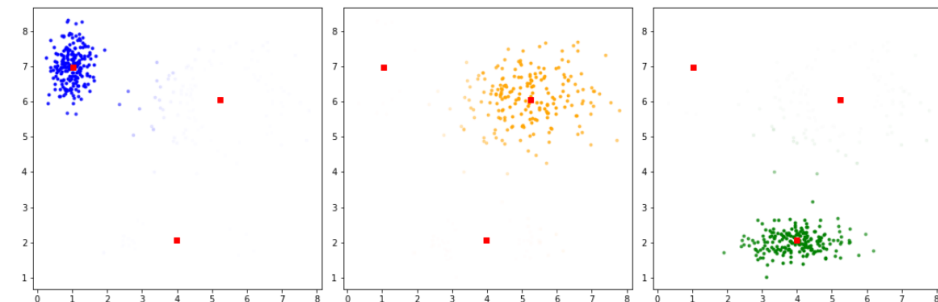
Iteration 6



Iteration 7



Iteration 8



Summary By Quick Quiz

✓ Four Types of Clusters

✓ Two Types of Clustering

✓ K-means Clustering

1. Centroid Initialization

2. Assignment (to nearest centroid)

3. Update

$$\mathbf{c}_k = \frac{\sum_{i=1}^N w_{ik} \mathbf{x}_i}{\sum_{i=1}^N w_{ik}}, k = 1, \dots, K$$

✓ Fuzzy C-means Clustering

1. Centroid Initialization

2. Assignment

$$w_{ik} = \frac{1}{\sum_{j=1}^C \left(\frac{\|\mathbf{x}_i - \mathbf{c}_k\|}{\|\mathbf{x}_i - \mathbf{c}_j\|} \right)^{\frac{2}{r-1}}}$$

3. Update

$$\mathbf{c}_k = \frac{\sum_{i=1}^N (w_{ik})^r \mathbf{x}_i}{\sum_{i=1}^N (w_{ik})^r}, k = 1, \dots, C$$

THE END